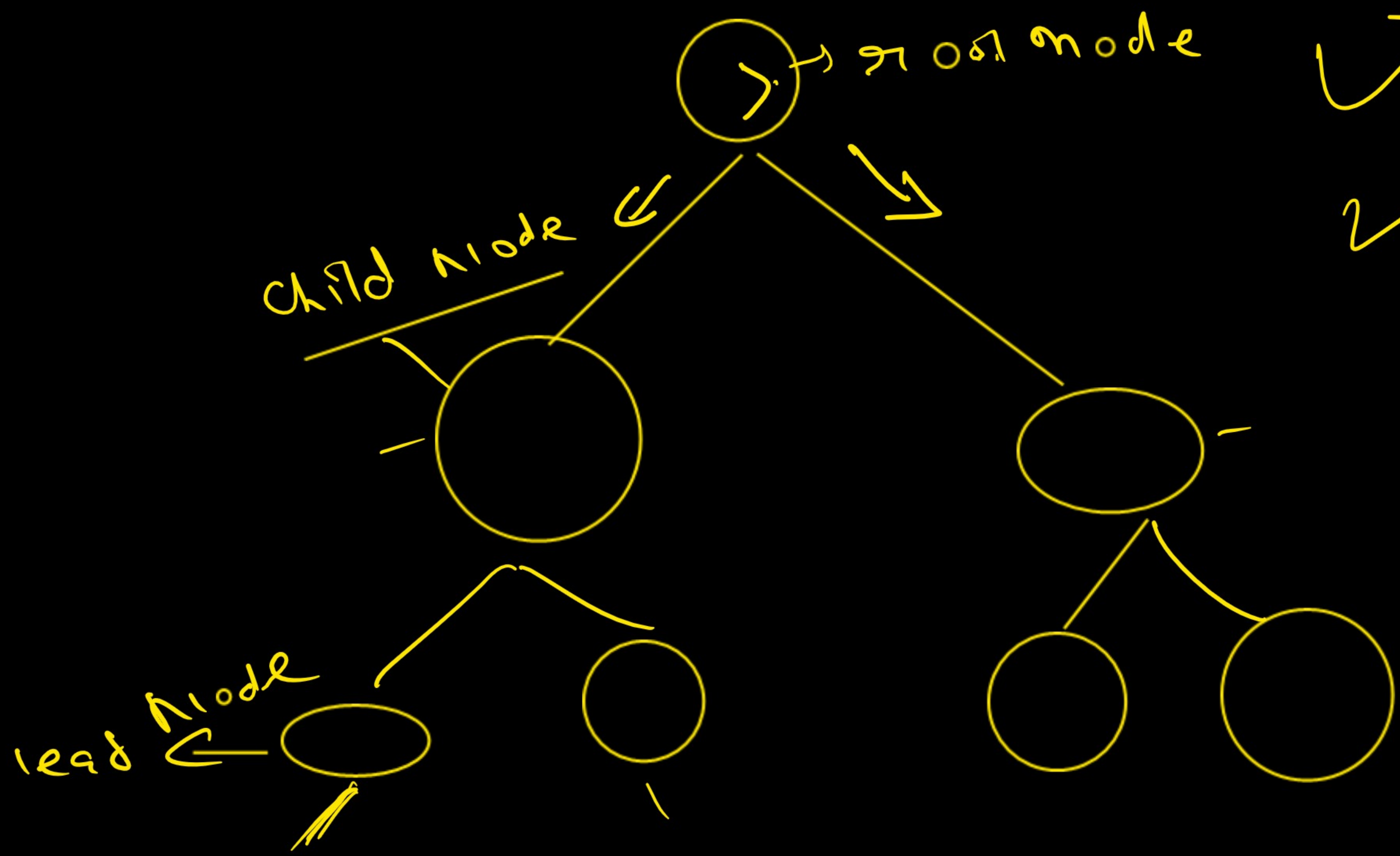
Decision tree



- Classification task
- Regression task

ID	Age	income(y)	y
A	25	30	0
B	70	35	0
C	28	45	0
D	32	55	1
E	38	60	1
F	45	65	0
G	48	70	0
H	29	58	1
I	31	27	?

Entropy + information Gain

$$E = -\sum_{i=1}^{N} p_i \log p_i \qquad p_0 = 5/8 \qquad p_1 = 3/8$$

$$E(S) = -\left(p_0 \log_2 p_0 + p_1 \log_2 p_1\right)$$

$$= -\left(\frac{5}{8}\log_2\frac{5}{8} + \frac{3}{8}\log_2\frac{3}{8}\right)$$

0.954434

income = 30, 35, 45, 55, 58, 60, 65, 70

$$t = \frac{45+55}{2} = 50$$

income

$S_L \le 50$ | S_R > 50

id	age	inc	y
A	25	30	0
B	30	35	0
C	28	45	0

id	age	inc	y
D	32	55	1
E	38	60	1
F	45	65	0
G	48	70	0
H	28	58	1

$$E(S_L) = -\frac{3}{3}\log_2\left(\frac{3}{3}\right)$$

$$= -\log_2(1)$$

$$= 0$$

$$E(S_R) = -\left(\frac{2}{5}\log_2\left(\frac{2}{5}\right) + \frac{3}{5}\log_2\left(\frac{3}{5}\right)\right)$$

$$= 0.9705$$

$$\text{Weighted Entropy} = \frac{|S_L|}{S} H(S_L) + \frac{|S_R|}{S} H(S_R)$$

$$= \frac{3}{8} \times 0 + \frac{5}{8} \times 0.9705 = 0.6068$$

$IG = H(S) - H_{a\,gder}$

$= 0.9544 - 0.6068$

$= 0.3475$

id	age	SR	inc	y
D	32	55		1
E	38	60		1
F	45	65		0
G	48	70		0
H	25	58		1

inc = 55, 58, 60, 65, 70

$t = \frac{58+60}{2} = 59$

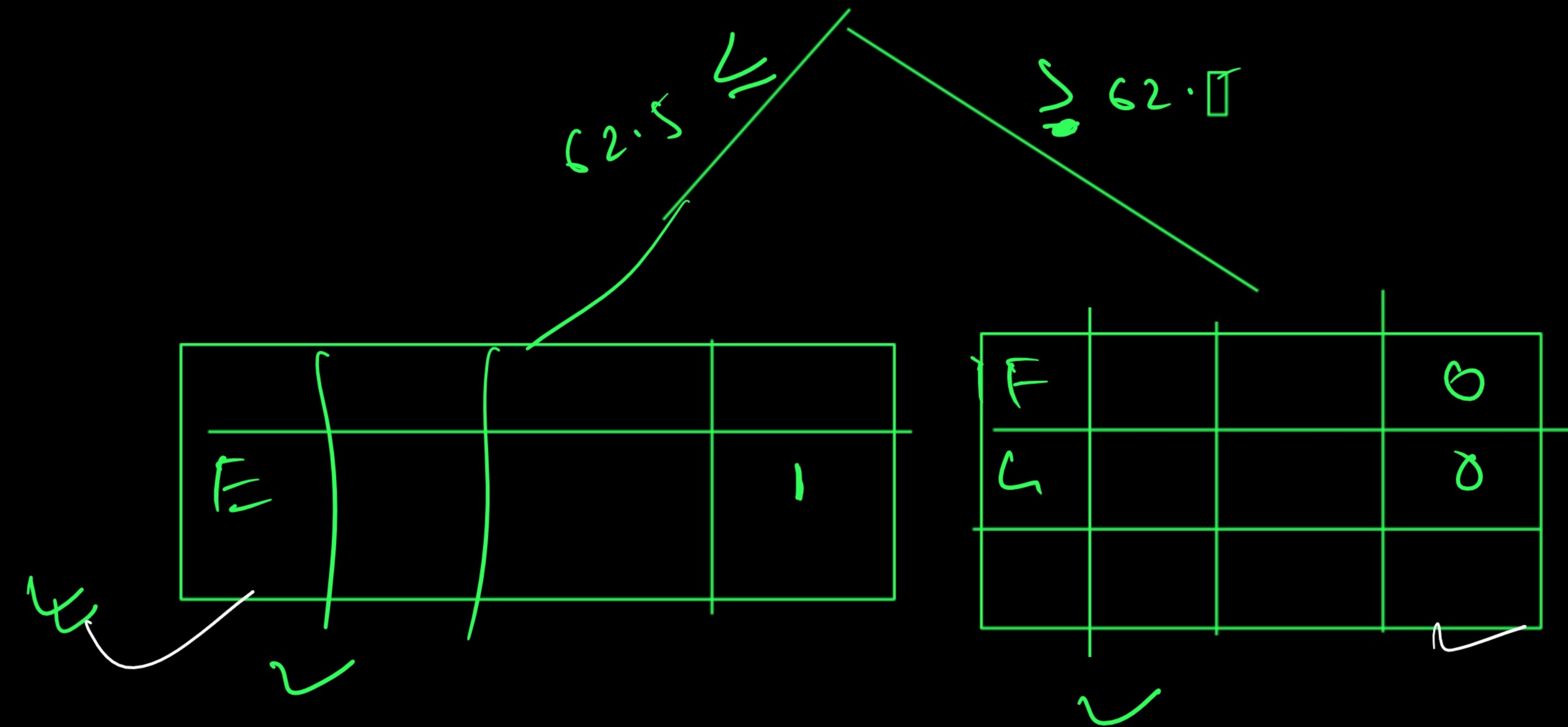
inc

$59 \le$

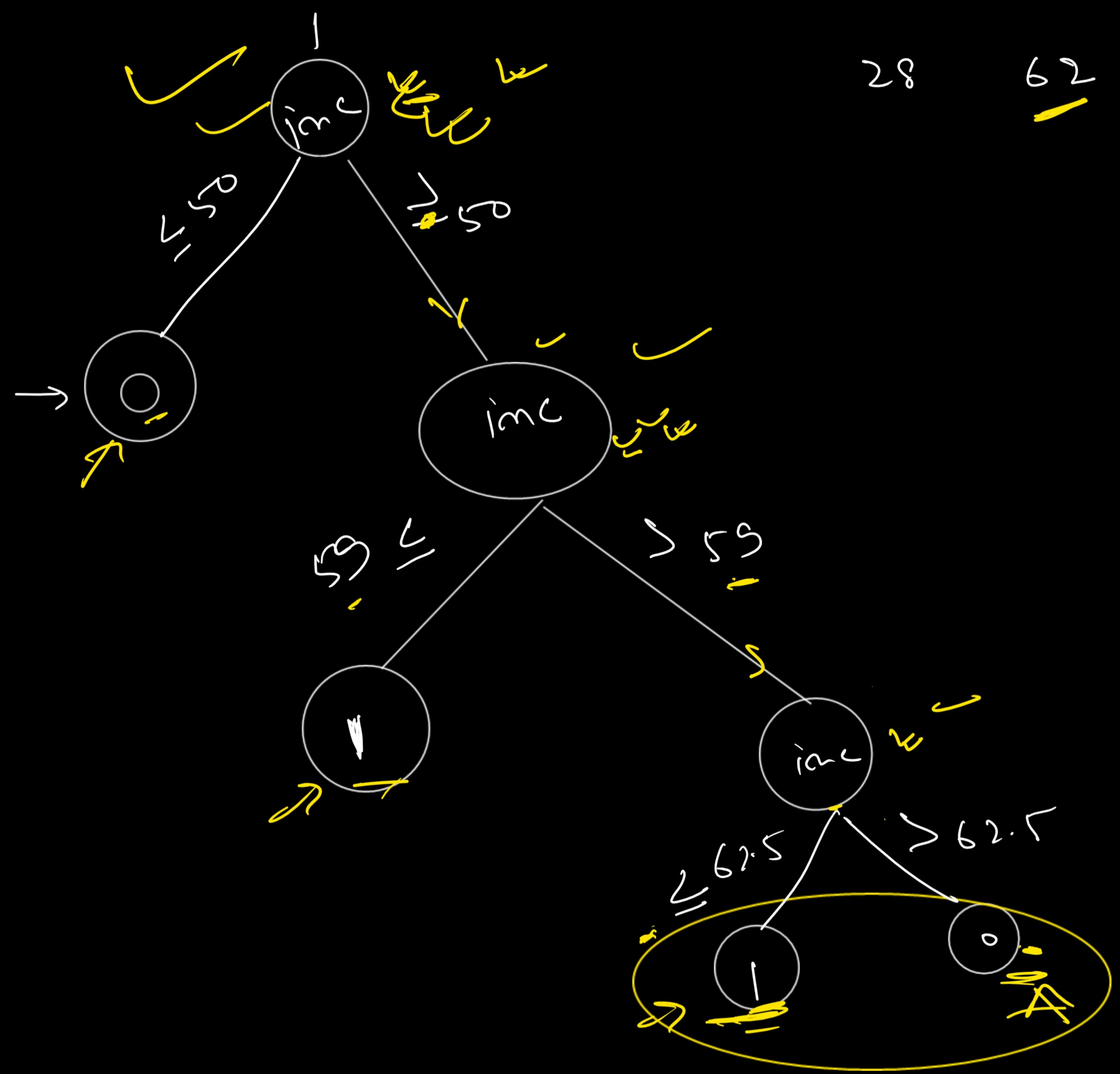
≥ 59

D		1
H		1

E	38	60	1
F	45	65	0
G	48	70	0



62.5
F
1
F
0
G
0



28 62 ?

$$\text{Gini} = \left(1 - \sum_i p_i^2\right)$$

$$= 1 - \left((p_0)^2 + (p_1)^2\right)$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$\Delta MSE = MSE_b - MSE_a$$

$$\hat{y}_{leaf} = \frac{1}{n} \sum (y_i)$$